Research landscapes and thematic evolution of acupuncture-related therapies for spasm and spasticity: a bibliometric and knowledge-mapping study

WORKING MANUSCRIPT — 11 October 2026

Editorial status: PubMed retrieval and an initial search audit have been completed. The Web of Science export, merged eligibility decisions, and CiteSpace analyses are outstanding. This document is a substantive writing draft, not a completed research report or a submission-ready manuscript. Planned procedures remain in the future tense; no network results have been invented.

Authors, affiliations, corresponding author and target journal: to be supplied by the research team.

Abstract

Background: Research on acupuncture-related interventions for spasm and spasticity spans different clinical conditions, intervention modalities and experimental models. A transparent bibliometric assessment requires consistent definitions, adequate retrieval and explicit handling of incomplete metadata.

Objective: To characterize publication patterns, collaboration networks, influential knowledge structures and thematic evolution in English-language research on acupuncture-related therapies for spasm and spasticity published through 10 October 2026.

Methods: The intended study combines Web of Science Core Collection and PubMed records after deduplication and metadata enrichment. Eligible publications will be research articles and reviews with an accessible abstract or full text and the required bibliographic fields. Planned analyses comprise annual output, country/institution/author collaboration, journal distribution, keyword co-occurrence, temporal evolution, clustering, three-field mapping and keyword bursts. Document co-citation will be assessed only where reference-list coverage is adequate. CiteSpace version assignments and Excel trend fitting will be reported after actual execution.

Results: Final bibliometric results are pending. In the completed PubMed search audit, the original strategy retrieved 395 records; an expanded sensitivity strategy retrieved 604, adding 209 candidate records. These are retrieval counts, not final inclusion counts. The near-topic 2026 bibliometric study identified during background searching was absent from the original strategy and present in the expanded strategy.

Conclusion: The search audit identifies material sensitivity and metadata-completeness issues that require resolution before the research landscape can be interpreted. Conclusions about leading contributors, thematic clusters or emerging hotspots cannot yet be drawn from the planned merged corpus.

Keywords: acupuncture; electroacupuncture; moxibustion; dry needling; spasm; spasticity; bibliometrics; CiteSpace.

1. Introduction

Spasticity is a form of increased muscle tone associated with upper motor neuron pathology, in which resistance changes with the velocity of passive stretch [1]. It is not interchangeable with every condition described as a spasm. A search encompassing both terms may retrieve neurological rehabilitation studies, hemifacial spasm research and investigations of visceral smooth-muscle contraction. This breadth can be valuable for mapping a field, but it also creates a methodological requirement: clinically different conditions must remain identifiable throughout screening, network construction and interpretation.

Acupuncture-related interventions have been investigated in several forms, including manual acupuncture, electroacupuncture, moxibustion and dry needling. Systematic reviews have examined acupuncture and electroacupuncture for post-stroke spasticity [2–5], while a randomized trial has evaluated dry needling in relation to spasticity, motor function and stability after stroke [6]. These publications provide distinct types of evidence and should not be treated as interchangeable indicators of treatment effectiveness. Their coexistence is relevant to a bibliometric study because a field can expand through clinical trials, evidence syntheses and methodological debate without a corresponding uniform increase in certainty about clinical benefit.

Experimental studies add another dimension. Research in a post-ischemic stroke rat model has investigated KCC2-mediated spinal GABAA signaling in association with acupuncture and altered reflex function [7]. Other investigations address different anatomical and clinical contexts, including colonic spasm during colonoscopy [8] and electroacupuncture for hemifacial spasm [9]. These examples motivate disease- and intervention-specific interpretation. A mechanism examined in an experimental stroke model cannot be assumed to explain all forms of spasm, and a bibliometric cluster connecting terms does not establish a biological pathway or a clinical indication.

Previous work has already combined bibliometric analysis with meta-analysis for acupuncture in post-stroke spastic hemiplegia [10]. The present study therefore does not claim to be the first bibliometric investigation of this general topic. Its proposed contribution is a more explicit account of the broader spasm/spasticity literature, with cross-database enrichment, transparent document-level screening and sensitivity analyses for terminology, metadata and network settings. Updating the end date alone would be insufficient to establish a substantial methodological contribution.

Science mapping offers complementary perspectives on this literature. Collaboration networks describe relationships among contributors; keyword networks organize recurring topics; and co-citation networks describe the reference structures shared by publications [11,12]. These relationships answer questions about the organization and development of research rather than the efficacy of an intervention. Their interpretation depends on the source corpus and on analytical choices such as term normalization, entity disambiguation, time slicing and pruning.

The study will address four questions: how publication output has changed over time; how research activity and collaboration are distributed across countries, institutions, authors and journals; which themes connect the different conditions and modalities; and which changes in keyword frequency are supported by reproducible temporal analyses. A search audit is necessary before addressing these questions because omissions at retrieval cannot be repaired by sophisticated visualization later in the workflow.

2. Materials and methods

2.1. Design, scope and reporting

This is a planned bibliometric and knowledge-mapping study of publications available from database inception through 10 October 2026. The current scope follows the supplied search strategy and encompasses spasm and spasticity without an a priori restriction to stroke. Clinical investigations, experimental studies and completed reviews will be considered. Disease categories will be recorded separately so that neurological spasticity is not conflated with hemifacial, visceral or other spasms. Search documentation will follow the transparency principles of PRISMA-S [13]; this does not imply that the study is a clinical-effectiveness systematic review.

The intended primary intervention categories are acupuncture, electroacupuncture, moxibustion, fire needling and dry needling. The final handling of protocols and adjacent modalities, including laser acupuncture and transcutaneous acupoint stimulation, must be settled before the eligible corpus is frozen. Such records are currently flagged rather than silently treated as completed studies or removed under an unstated rule.

2.2. Databases and search strategy

PubMed was searched using the supplied combination of an acupuncture concept and a spasm/spasticity concept. The executable strategy and NCBI query translation have been archived with the raw response [14]. A publication-date restriction ended on 10 October 2026. Language and publication-type filters were not applied at retrieval so that their subsequent exclusions could be documented. The complete original PubMed result set was downloaded in XML batches, and the retrieved identifiers were checked against the search result list.

The intended second database is Web of Science Core Collection. Its precise citation indexes, accessible time coverage, search date, result count and full search history will be documented when the export is obtained. The topic query is TS=((acupuncture OR electroacupuncture OR moxibustion OR fire needle OR dry needle) AND (spasm OR spasticity)). Full records and cited references will be exported in tagged plain text [15]. Records will not be described as retrieved from Web of Science until this export is available and verified.

A separate PubMed sensitivity search added the Muscle Spasticity MeSH heading, spasm* and spastic* word forms, the electro-acupuncture spelling variant, and dry needl* and fire needl* phrases. Its identifiers have been retained separately. Additional retrieval is not assumed to represent additional eligible literature. If this expanded strategy is adopted as the principal strategy, the Web of Science search will also be revised and the complete screening process repeated under the revised search version.

Search execution dates and publication eligibility dates will be reported separately. The current retrieval is not a reconstruction of the database as it existed on the cutoff date. For records with conflicting electronic and print dates, the earliest verifiable public publication date will determine cutoff eligibility, while the annual-output year rule will be applied consistently and documented. Records with insufficiently precise dates near the cutoff will require verification.

2.3. Eligibility and screening

Eligible publications must be in English, directly relevant to an in-scope intervention for spasm or spasticity, and accessible as an abstract or full text. Research articles and reviews will be considered. Duplicate records, duplicate publications, conference abstracts, individual case reports, reports, notices, letters and editorial material will be excluded. Retraction notices, errata and retracted articles will be identified separately. The treatment of analytical case series will be based on study design rather than title wording alone.

Records lacking a publication year, author names, institutional information or keywords will be excluded only after attempted cross-database enrichment and source verification. An absent PubMed abstract will trigger a full-text availability check rather than immediate exclusion. English-language abstracts accompanying non-English articles will not satisfy the publication-language criterion by themselves.

The screening ledger preserves raw publication-type tags, multiple diagnostic flags, one primary exclusion reason and a reasoned relevance decision. The proposed hierarchical accounting order is duplicates, ineligible publication type, non-English language, ineligible publication date, unrelated content, inaccessible abstract/full text, and unresolved mandatory metadata. Counts of case reports, editorials, letters and unrelated publications will be reported individually. Two independent researcher assessments and adjudication are recommended for final decisions; they have not yet been performed. The current ledger records partial AI-assisted triage and leaves researcher-review fields blank.

2.4. Deduplication and data harmonization

DOIs, PMIDs and Web of Science accession numbers will be normalized before matching. Identical identifiers will generate high-confidence match candidates; conflicting identifiers or materially different titles will be reviewed. Normalized title-plus-year matches will also be reviewed rather than merged indiscriminately. A protocol and a subsequent results article will remain distinct publications unless the final eligibility policy specifies otherwise.

Merged records will retain source identifiers and field provenance. Missing author keywords or affiliations may be supplemented from the corresponding record in the other database or verified source. Author keywords, MeSH headings and Keywords Plus will remain separate data types. The primary keyword analysis will use author keywords, with alternative term sources examined separately if appropriate. An explicit dictionary will preserve raw terms and their normalized equivalents.

Countries will be assigned from author affiliations, not from journal publication locations or author names. Institution harmonization will distinguish departments, hospitals and universities before any aggregation. Author disambiguation will use full names, affiliations, coauthorship patterns and identifiers where available. Within each paper, repeated occurrences of an entity will count once. Country, institution and author productivity will use full counting; their totals can consequently exceed the number of papers.

2.5. Descriptive statistics and annual publication trends

Annual output will be calculated from the final deduplicated eligible corpus, including intervening years with zero publications. A prespecified ordinary least-squares linear model of annual counts against publication year will describe the complete-year series ending in 2025. The fitted equation, interval and coefficient of determination will be reported. Publications from 2026 will be plotted separately as an incomplete year through 10 October and excluded from the primary regression. R^2 will be treated as a descriptive measure of fit, not as evidence of causation or a test of clinical progress.

The requested Excel implementation remains planned. The actual Excel version and executed calculations must be recorded before the manuscript states that Excel was used. Rankings will identify the ten most productive countries, institutions, authors and source journals and the ten most frequent keywords, with an explicit rule for ties. No rankings will be calculated from an incompletely screened retrieval set and presented as final results.

2.6. Network construction and visualization

Country, institution and author networks will represent coauthorship relationships. Keyword edges will represent co-occurrence within papers, and document co-citation edges will represent references cited together by source publications. Reference availability will be audited by database of origin. PubMed-only records lacking complete reference lists will not be treated as papers with zero references; a

restricted co-citation subset may be necessary and must be reported separately from the descriptive corpus.

Node size will encode the specified publication, keyword or citation frequency. Edge width will encode the defined relationship weight. Colors will be identified in each legend as cluster membership or temporal information. Distances produced by a layout algorithm will not be described as direct numerical measures of association. A country map will combine geographic locations and collaboration links; its base-map source and any additional plotting software will be documented.

The requested journal table requires an explicit definition of connection strength. The working proposal is source-journal bibliographic coupling, aggregating shared-reference links between papers published in different journals. This is distinct from cited-journal co-citation strength. The selected definition, reference coverage and counting convention will be frozen before this column is populated.

2.7. CiteSpace settings, clusters and bursts

CiteSpace is planned as the principal science-mapping application [11,12]. The requested assignment is version 6.2.R6 Advanced for the main analyses, 6.2.R7 for Figure 5 and 6.2.R3 for Figure 10. These are requested versions, not verified execution history. Installation, licensing availability, exact builds and run logs must be recorded before any completed-use statement is made.

The initial configuration proposes one-year slices and g-index selection with k=25, with k=15 and k=50 examined for sensitivity. Original unpruned networks and visualization-pruned versions will both be retained. Pathfinder and pruning of sliced networks are proposed visualization settings; the specific network used to calculate centrality and clustering must be identified. Betweenness centrality will describe bridging positions, and the display convention for highlighting nodes above 0.1 will be checked in the actual run.

Cluster characterization will report modularity Q, cluster size, mean publication year and silhouette values. The commonly used Q and silhouette cutoffs will be interpreted as heuristic descriptors rather than statistical significance tests. Small clusters with high silhouette values will be examined cautiously. LLR and LSI labels derived from titles will be exported separately and interpreted by checking representative publications; automated labels will not substitute for content review.

Keyword burst analysis will detect changes in occurrence frequency, not changes in citation counts. The provisional starting settings are gamma=1.0 and a minimum burst duration of two years, with alternative gamma values assessed if supported by the installed version. Burst strength, onset and end years, the actual parameter settings and the handling of the incomplete final year will be retained. A burst reaching 2026 will be treated as right-censored rather than as a trend known to have ended.

2.8. Temporal and three-field analyses

Keyword temporal evolution will be examined using annual occurrence tables and corresponding timeline views. The three-field diagram will link authors, the ten most frequent normalized keywords, and source journals. For a paper containing a authors and m selected keywords, each author–keyword link will receive weight 1/(a×m), and each keyword–journal link will receive weight 1/m. Thus, each represented paper contributes unit mass to both stages. Any display restriction on authors or journals and aggregation into an Other category will be stated explicitly. The actual software used for this diagram will be reported in addition to CiteSpace and Excel.

3. Results — current evidence and outstanding analyses

3.1. Completed retrieval and preliminary screening audit

The original PubMed query retrieved 395 unique PMIDs, comprising 394 journal records and one NICE evidence-review book/report record. All retrieved identifiers were represented in the downloaded XML. Of the 395 records, 257 carried an English-language designation. Among the 394 journal records, 186 lacked author keywords and 50 lacked affiliation information. These are overlapping pre-enrichment flags, not final exclusion counts.

The PubMed tags identified 49 case reports, one letter and no Editorial-tagged records. An initial examination of titles and selected abstracts identified one additional two-patient case report, one additional letter and one retraction notice. Ten records were provisionally classified as unrelated to the intended treatment question; the individual reasons are preserved in the screening ledger. Under the current hierarchical accounting, 202 records have provisional exclusion reasons and 193 remain unresolved or provisionally retained. These counts must not be interpreted as the final study flow.

The sensitivity search retrieved 604 records, including all 395 original records and 209 additional identifiers. The nearby 2026 bibliometric study [10] was found by the sensitivity strategy but not by the original strategy. This observation supports further retrieval validation; it does not establish the eligibility of every additional record. Both result sets are archived separately.

3.2. Final study-selection and publication results

Awaiting the Web of Science export, deduplication, completed relevance review and post-enrichment eligibility decisions. Figure 1 will report the complete flow. Figure 2 will present annual output and the complete-year regression. The final sample size, study period, annual peak, fitted equation and R^2 are deliberately not supplied in this draft.

3.3. Contributors, sources and thematic structure

Awaiting computation from the frozen eligible corpus. Country, institution and author results will appear in Figures 3–5 and Tables 1–3. Source-journal productivity and the defined connection strength will appear in Table 4. Keyword co-occurrence, temporal evolution, clustering, the three-field diagram and bursts will appear in Figures 6–10 and Tables 5–6. Where reference coverage permits, Supplementary Figure S1 will show document co-citation. No leading entity, centrality value, cluster label, Q, silhouette value or burst strength has been inferred from background knowledge.

4. Discussion — interpretive framework pending final results

4.1. Search scope is part of the scientific result

The completed audit shows that a seemingly small difference in terminology can materially change retrieval. The additional candidates and the near-topic missed record justify checking recall before interpreting publication trends. Conversely, broad retrieval can introduce studies in which acupuncture or spasm appears only as background, a prior treatment, a sensory examination term or a surgical indication. Transparent document-level reasons are therefore necessary to distinguish search yield from the actual subject literature.

The distinction between spasm and spasticity should remain visible in the final analysis. If a large stroke-related cluster emerges, it should be interpreted alongside the number and composition of non-stroke publications. A small visceral-spasm cluster could reflect a different specialty and intervention context rather than an immature version of neurological rehabilitation research. These are questions to be tested against the completed maps, not conclusions already supported by the current retrieval set.

4.2. Bibliometric prominence and clinical evidence

The existence of multiple systematic reviews and primary studies motivates examination of how evidence types interact [2–9]. A frequently occurring intervention or highly connected institution may indicate sustained research activity, but neither proves therapeutic effectiveness. Mechanistic publications may bridge molecular and rehabilitation terminology without establishing translation from animals to patients. Any discussion of clinical implications must therefore remain separate from network prominence and refer directly to the relevant clinical evidence.

If the final keyword analysis identifies themes concerning inhibitory signaling, brain plasticity or combined rehabilitation, interpretation should check the publications contributing those terms. Automated cluster labels may privilege a small set of titles or a newly popular phrase. LLR and LSI agreement, representative-record inspection and stability across settings will be more informative than naming visually appealing clusters alone.

4.3. Metadata-related selection and other limitations

The requirement for complete affiliation and keyword fields can create selection bias, particularly if older publications are less completely indexed. The final report should quantify missingness before and after enrichment and compare the time distribution of excluded and retained records. Where useful, an ancillary descriptive analysis can retain otherwise relevant records lacking keywords, while the prespecified main analysis continues to follow the requested complete-metadata criterion. Any such ancillary analysis must be clearly labeled and not used to change eligibility retrospectively without documentation.

English-language restriction, database coverage, indexing delay and reference-list availability will constrain generalizability. Author name ambiguity and institutional name changes may distort collaboration estimates. Different network thresholds, pruning settings and software builds may alter centrality and cluster structure. The incomplete 2026 observation window also limits temporal comparisons and burst interpretation. Finally, merging databases improves coverage but does not make their citation metadata uniform; co-citation findings may apply to a narrower corpus than publication counts.

5. Conclusion

A reproducible PubMed retrieval and search audit have been completed, and the analytical framework for a combined PubMed–Web of Science study has been specified. The current evidence identifies a need for search-sensitivity validation and metadata enrichment. Conclusions about the structure, leading contributors or emerging themes of acupuncture-related research on spasm and spasticity remain pending completion of the merged-corpus analysis.

Declarations and availability

Funding, competing interests, author contributions and acknowledgments: to be supplied and verified by the authors. No authorship or funding statement has been invented.

Ethics: the proposed analysis uses published bibliographic information and does not recruit participants. Any statement about formal exemption or institutional review must reflect the applicable institutional requirements; no approval or exemption number has been assigned here.

Data and code: the local project contains original PubMed search responses, XML batches, checksums, a record-level screening ledger, supplementary sensitivity identifiers and reproducible parsing/audit scripts. Web of Science records and a final analytic dataset are not yet available. Public redistribution of abstracts and subscription-database records must follow the corresponding licenses. No repository deposition has been claimed.

AI assistance: AI was used for retrieval scripting, initial record triage, methodological checks and drafting. The authors must verify eligibility decisions, interpretations and citations and make any disclosure required by the target journal. The present workflow does not constitute two independent human reviews.

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